

1 The off-line counterfactual inside the original holonomy construction

This calculation uses the original theta source in the final analytic continuation of the programme. In particular, the scalar measure below is computed from g/h with $g = 2\xi$; it is not a Gamma reference measure. The source equations being continued are (H5)–(H17), (H24)–(H28), and (H53)–(H55) of *All-holonomy descent of the original theta quotient*. The complete source record and locators are recorded in the accompanying audit. The conclusions concern the stated counterfactual and do not assert the existence of an off-line zero.

1.1 The retained arithmetic objects and every phase

Suppose $1/2 + \delta + i\gamma$ is a zero of g of its full order m , where $0 < \delta < 1/2$ and $\gamma > 0$. Reflection and conjugation give the complete quartet, with the same order. Retain

$$h(s) = \prod_{\epsilon, \eta \in \{-1, 1\}} (s - 1/2 - \epsilon\delta - i\eta\gamma)^m, \quad v_h = g/h, \quad w_h(u) = \frac{|v_h(1/2 + iu)|^2}{2\pi}. \quad (\text{HT1})$$

The division is entire and its full jet $v_h = j_h(v_h)$ is a unit. For an integer $k \geq 2$, put

$$c = k/2, \quad \ell_k = 1 + k(m-1), \quad q = \ell_k(k+1)^2, \quad m_{h,k} = w_h^{*k},$$

$$\chi(S) = \prod_{a,b=0}^k (S - c - (2a-k)\delta - i(2b-k)\gamma)^{\ell_k}, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S. \quad (\text{HT2})$$

These are the original cyclic sum quotient, its full multiplicities, and its multiplication operator. For completeness, in each tensor local factor the nilpotent sum is $n_1 + \dots + n_k$ with $n_i^m = 0$. Its power $k(m-1)$ has nonzero coefficient $(k(m-1))!/((m-1)!)^k$ on $n_1^{m-1} \dots n_k^{m-1}$, whereas the next power vanishes. Thus its nilpotency index is exactly ℓ_k . Every pair (a, b) is realized by choosing independently the real and imaginary signs in the k tensor factors. Distinct pairs give distinct complex sums, so the annihilator in the sum variable is exactly χ .

The original analytic construction provides F_h with Mellin transform v_h , and the original source and jet maps are

$$\mathcal{V}P = P(D_1 + \dots + D_k)F_h^{\otimes k}, \quad J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad \eta[P] = v_h^{\otimes k}P(A_k)1. \quad (\text{HT3})$$

Here η is an injective complex linear module map. Its unit factor is retained. The norm of this source is exactly $\int_{\mathbb{R}} |P(c + iu)|^2 m_{h,k}(u) du$. The half-density map is

$$(\mathcal{U}_k F)(r, z) = e^{(kr + \sum_{i < k} z_i)/2} F(e^{r+z_1}, \dots, e^{r+z_{k-1}}, e^r), \quad r = \log x_k, \quad z_i = \log x_i - \log x_k. \quad (\text{HT4})$$

It maps into $L^2(\mathbb{R}, dr; \mathcal{K})$ with $\mathcal{K} = L^2(\mathbb{R}^{k-1}, dz)$ and intertwines $D_1 + \dots + D_k$ with $-\partial_r + c$. No relative variable is removed.

Fix any $L > 0$ and $\theta \in [0, 2\pi)$. The source of phase θ is

$$(\mathcal{Z}_{L,\theta}\psi)(r) = \sum_{j \in \mathbb{Z}} e^{ij\theta} \psi(r + jL), \quad \psi(r + L) = e^{-i\theta} \psi(r) \quad \text{in the target domain.} \quad (\text{HT5})$$

On the original finite source, its Gram form is exactly the polynomial norm for the measure

$$\nu_{L,\theta} = \frac{2\pi}{L} \sum_{n \in \mathbb{Z}} m_{h,k}(u_n) \delta_{u_n}, \quad u_n = \frac{2\pi n + \theta}{L}, \quad S = c + iu. \quad (\text{HT6})$$

The factor $2\pi/L$ is unchanged. The measure has positive mass at every u_n . Indeed the nonzero entire function v_h has only isolated zeros on the line; hence $w_h > 0$ almost everywhere. For every u , the product $w_h(v)w_h(u-v)$ is positive almost everywhere in v , so its integral is positive. Iterated convolution preserves strict positivity. The original weighted Schwartz estimates give convergence and all moments in (HT6). A nonzero polynomial cannot vanish at every lattice point; consequently every finite polynomial Gram matrix is positive definite.

For $N \geq q-1$, let $P_N = \mathbb{C}[S]_{\leq N}$ and $J_N = \pi_\chi|_{P_N}$. With the fixed coefficient bases define

$$M_{N,\theta} = (\mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V}|_{P_N})^*(\mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V}|_{P_N}), \quad K = J_N M_{N,\theta}^{-1} J_N^*, \quad G = K^{-1},$$

$$C_{N,\theta} = M_{N,\theta}^{-1} J_N^* G, \quad R = \mathcal{Z}_{L,\theta}\mathcal{U}_k\mathcal{V} C_{N,\theta}. \quad (\text{HT7})$$

Thus $J_N C_{N,\theta} = I_E$ and $R^* R = G$. All inverses in (HT7) exist: $M_{N,\theta}$ is positive definite and J_N is surjective. No inverse at degree $q-2$ is used.

1.2 A uniform phasewise lower bound for the original control

Define

$$H = G^{-1/2}(A^*G + GA - kG)G^{-1/2}, \quad \mathcal{L}_{h,k} = 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{HT8})$$

For every $k \geq 2$, $L > 0$, $\theta \in [0, 2\pi)$, and $N \geq q-1$, the nonzero eigenvalues of H are $\varepsilon, -\varepsilon$, and

$$\boxed{\varepsilon \geq \mathcal{L}_{h,k} > 0, \quad \frac{\mathcal{L}_{h,k}}{q} \geq \frac{\delta k}{2}.} \quad (\text{HT9})$$

Proof. Let $p_j(S)$ be the monic orthogonal polynomials for (HT6), with squared norms $\omega_j > 0$, and let $b_j = [p_j] \in E$. In the real coordinate u , write $p_j(c + iu) = i^j r_j(u)$, where r_j is real monic. Orthogonality and the symmetry of multiplication by u show

$$ur_j = r_{j+1} + a_j r_j + \frac{\omega_j}{\omega_{j-1}} r_{j-1}, \quad a_j \in \mathbb{R},$$

with the last term omitted at $j=0$. To verify that no earlier term occurs, pair ur_j with r_i , $i \leq j-2$, and move u to r_i ; its degree is at most $j-1$. Pairing with r_{j-1} gives precisely ω_j/ω_{j-1} . In the S coordinate this is

$$Sp_j = p_{j+1} + (c + ia_j)p_j - \frac{\omega_j}{\omega_{j-1}} p_{j-1}. \quad (\text{HT10})$$

Since the source Gram is diagonal in this basis, $K = \sum_{j=0}^N b_j b_j^* / \omega_j$. Apply A on the left and A^* on the right and use (HT10). The imaginary diagonal terms cancel, and each adjacent term except the last cancels its predecessor. This gives, including $N = q-1$,

$$AK + KA^* - kK = \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N}. \quad (\text{HT11})$$

Congruence by $G^{1/2}$ identifies the left side with H . Thus its rank is at most two. Also $\text{Tr } H = 2 \text{Re } \text{Tr } A - kq = 0$ by the complete grid (HT2).

Let E_+ be the direct sum of the complete primary spaces with $a > k/2$, and let P be the orthogonal projection onto $G^{1/2}E_+$ in the standard Hermitian space. The latter subspace is invariant under $G^{1/2}AG^{-1/2}$. Taking traces in an orthonormal basis of this subspace gives

$$\text{Tr}(PH) = 2 \sum_{a > k/2} \sum_{b=0}^k \ell_k(2a-k)\delta = 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor = \mathcal{L}_{h,k}. \quad (\text{HT12})$$

The finite sum is r^2 if $k = 2r - 1$, and $r(r + 1)$ if $k = 2r$. A Hermitian rank-at-most-two matrix of trace zero has eigenvalues $\varepsilon, -\varepsilon$ and zeroes. For every orthogonal projection, $\text{Tr}(PH) \leq \varepsilon$: in an eigenbasis the positive contribution is at most ε and the negative one is nonpositive. Equation (HT12) proves the first inequality of (HT9), and $4\lfloor(k + 1)^2/4\rfloor \geq k(k + 1)$ proves the second. $\square$

1.3 The exact energy forced into the retained theta boundary

In $\mathcal{H}_{L,\theta} = L^2([0, L], dr; \mathcal{K})$ put $D_{L,\theta} = -\partial_r + c$, with domain $\{f \in H^1([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0)\}$. The domain of its square is $\{f \in H^2([0, L]; \mathcal{K}) : f(L) = e^{-i\theta} f(0), \quad f'(L) = e^{-i\theta} f'(0)\}$. The original smooth periodized source columns belong to these domains. Monic division in the original coefficient source gives the unique polynomial map $Q_{N,\theta}$ with

$$SC_{N,\theta} - C_{N,\theta}A = \chi Q_{N,\theta}.$$

Indeed applying J_{N+1} gives zero. Define the actual boundary

$$\mathcal{B} = \mathcal{Z}_{L,\theta} \mathcal{U}_k \mathcal{V}(\chi Q_{N,\theta}) = D_{L,\theta} R - RA. \quad (\text{HT13})$$

This retains its theta primitive. Explicitly, tensor monic division gives

$$\chi(S)Q(S) = \sum_i h(s_i)Q_i(\mathbf{s}),$$

because its image in the full tensor jet algebra vanishes. In the original theta complex, ϕ_h has degree zero, F_h has degree one, and $h(D)F_h = \Theta\phi_h$. The degree- $(k - 1)$ tensor primitive is exactly

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) (F_h^{\otimes(i-1)} \otimes \phi_h \otimes F_h^{\otimes(k-i)}).$$

The tensor differential contributes a further sign $(-1)^{i-1}$ when it acts on ϕ_h , because the preceding $i - 1$ factors have degree one. The two signs cancel, so its differential is precisely $\mathcal{V}(\chi Q)$. Applying the same half-density and phase maps gives precisely (HT13).

Set

$$U = RG^{-1/2}, \quad T = \mathcal{B}G^{-1/2}, \quad Z = U^*T, \quad T_\perp = (I - UU^*)T.$$

Then the following exact energy identity, with no omitted source term, holds:

$$\boxed{\text{Tr}(G^{-1}\mathcal{B}^*\mathcal{B}) = \frac{\varepsilon^2}{2} + \left\| \frac{Z - Z^*}{2} \right\|_{\text{HS}}^2 + \|T_\perp\|_{\text{HS}}^2 \geq \frac{\mathcal{L}_{h,k}^2}{2} \geq \frac{\delta^2 k^2 q^2}{8}.} \quad (\text{HT14})$$

All quantities on the left are calculated from g/h and its original theta representatives. In particular the same estimate holds after integration with the original Fourier inversion measure:

$$\int_0^{2\pi} \text{Tr}(G_{N,\theta}^{-1} \mathcal{B}_{N,\theta}^* \mathcal{B}_{N,\theta}) \frac{d\theta}{2\pi} \geq \frac{\mathcal{L}_{h,k}^2}{2}. \quad (\text{HT15})$$

Proof. The two arguments in integration by parts have the same boundary multiplier $e^{-i\theta}$, so the endpoint products cancel. Hence $D_{L,\theta}^* + D_{L,\theta} = 2cI$ on the stated domain. Inserting $D_{L,\theta}R = RA + \mathcal{B}$ and $R^*R = G$ gives

$$A^*G + GA - kG = -(R^*\mathcal{B} + \mathcal{B}^*R), \quad H = -Z - Z^*.$$

Since $U^*U = I$, the decomposition $T = UZ + T_\perp$ is orthogonal in Hilbert–Schmidt norm. The Hermitian and skew-Hermitian parts of Z are orthogonal for the real Hilbert–Schmidt scalar product. Therefore

$$\|T\|_{\text{HS}}^2 = \frac{1}{4}\|H\|_{\text{HS}}^2 + \left\|\frac{Z - Z^*}{2}\right\|_{\text{HS}}^2 + \|T_\perp\|_{\text{HS}}^2.$$

By (HT9), $\|H\|_{\text{HS}}^2 = 2\varepsilon^2$, which proves (HT14). For fixed k, L, N the weighted source sums and their derivatives converge uniformly in θ . Thus $M_{N,\theta}$ is continuous. Its smallest eigenvalue is positive on the compact phase circle, so its inverse, $G_{N,\theta}$, and the boundary columns are continuous there. Integration of (HT14) proves (HT15). $\square$

1.4 The full lattice test for the original Laplacian boundary

Put $\mathcal{L}_{L,\theta} = D_{L,\theta}^2 - kD_{L,\theta} = \partial_r^2 - c^2$. Direct substitution of (HT13) gives the exact map

$$\mathcal{L}_{L,\theta}R - R(A^2 - kA) = (D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A. \quad (\text{HT16})$$

If $Av = \lambda v$ with $v \neq 0$, write $\lambda = c + x + iy$. Then

$$\frac{\|((D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A)v\|}{\|Rv\|} \geq \min_{n \in \mathbb{Z}} \sqrt{(u_n^2 + x^2 - y^2)^2 + 4x^2y^2}. \quad (\text{HT17})$$

In particular every phase and every admitted degree satisfy, on an eigenvector for $\lambda = k(1/2 + \delta + i\gamma)$,

$$\boxed{\frac{\|((D_{L,\theta} - k)\mathcal{B} + \mathcal{B}A)v\|}{\|Rv\|} \geq 2k^2\delta\gamma.} \quad (\text{HT18})$$

For an eigenvalue with $x \neq 0$ and $y = 0$, (HT17) also gives the positive lower bound $x^2 + \min_n u_n^2$, even though its imaginary part is zero.

Proof. The Fourier basis $L^{-1/2}e^{-iu_nr}$ diagonalizes the quasi-periodic Laplacian with eigenvalues $-u_n^2 - c^2$, each acting identically on $\mathcal{K}$. The source vectors lie in its domain by the original weighted differentiability estimates. Writing Rv in that basis and using Parseval gives

$$\|(\mathcal{L}_{L,\theta} - (\lambda^2 - k\lambda))Rv\|^2 = \sum_n ((u_n^2 + x^2 - y^2)^2 + 4x^2y^2) \|\widehat{Rv}(n)\|_{\mathcal{K}}^2.$$

The lattice is discrete and the expression tends to infinity as $|n| \rightarrow \infty$, so its infimum is a minimum. Also R is injective because $R^*R = G \succ 0$. Divide the displayed identity by $\|Rv\|^2$; (HT16) proves (HT17). The extreme primary factor in (HT2) is nonzero and has an eigenvector, so substituting $x = k\delta, y = k\gamma$ proves (HT18). If $y = 0$, the square root in (HT17) is $u_n^2 + x^2$. $\square$

1.5 What these computations decide

The original off-line counterfactual survives passage to every finite holonomy quotient only with the explicit boundary energy (HT14) and Laplacian residual (HT17). These quantities belong to the original arithmetic source. The statement is compatible with the exact all-phase identity $\int M_{N,\theta} d\theta / (2\pi) = M_N$ and with the original small gluing-norm estimate. That estimate concerns $C_N - C_{N,\theta}$; (HT13) applies S and subtracts right multiplication by A , preserving a different, explicitly displayed source quantity. There is no claim here that averaging the $G_{N,\theta}$ preserves the rank-two identity (HT11): the lower bound (HT15) integrates the proved phasewise energies themselves.

The transcript's threshold-four volume lower bound was already present before its final Deligne comparison. Its last counterexamples retain many generic identities but do not satisfy the actual divisor condition $h \mid 2\xi$. Consequently they disprove the reduced generic inference stated there; they do not disprove the entire arithmetic programme or provide an RH counterexample. Equations (HT9), (HT14), and (HT17) give further quantified consequences within the original programme without replacing its theta metric or its represented relations.

2 Actual bounds for the retained four-window costs

Continue with (HT1)–(HT18), now with $k \geq 3$. All unadorned source norms and quotient volumes in this section are those of the original $m_{h,k}(u) du$, with $g = 2\xi$ and the complete hypothetical quartet. Write $\mathcal{L} = \mathcal{L}_{h,k}$ and retain $q = \ell_k(k+1)^2$. The threshold-four lower bound is inherited from *Consecutive identities and the first window*, (CJ.9)–(CJ.19); it is already explicitly present in the recovered transcript at lines 4202–4228. It is not a new claim of the present continuation.

2.1 Every term of the inherited lower bound

Put $d_j = V_j/V_{j-1}$ for $j \geq q$. For the two original windows $I_- = (q-1, 2q-1)$ and $I_+ = (q, 2q)$, retain the contraction penalties

$$\begin{aligned} P_- &= -\log(1 - d_{2q-1}) - 2 \sum_{j=q}^{2q-2} \log(1 - d_j), \\ P_+ &= -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{j=q+1}^{2q-1} \log(1 - d_j). \end{aligned} \tag{HC1}$$

The source identities prove $0 < d_j < 1$ in these ranges. The real phase ϕ_N is defined by the retained source pairing $b_N^* G_N b_{N+1} = i\omega_N \phi_N$. The first window uses its original boundary identity; no inverse at degree $q-2$ occurs. Define weights

$$w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \quad (q \leq N \leq 2q-2), \quad \sum_{N=q-1}^{2q-1} w_N = 2q.$$

Adding the two exact source products gives

$$\mathcal{B}_k := \log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} = \sum_{N=q-1}^{2q-1} w_N \log(\varepsilon_N^2 + \phi_N^2) - \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q} + P_- + P_+. \tag{HC2}$$

Here every $\varepsilon_N \geq \mathcal{L}$, including the first degree.

For clarity, retain all constants in the inherited comparable-window bound. With $\alpha = \pi/2$, $B = 42 + 8m$, and $\vartheta_h = \int_{-1}^1 w_h$, put

$$\begin{aligned} t_h &= \min(1, \vartheta_h), \quad c_0 = \min(1, 2c_h/3), \quad \ell_h = \tfrac{1}{2} e^{-\alpha} \sqrt{c_0 t_h} 2^{-B/2}, \\ X &= \max(1, M_h(b), M_h(-b)), \quad U_h = 4\sqrt{X}/b, \\ C_h^{\text{win}} &= 27 \max(1, U_h)^3 \max(1, \ell_h^{-1})^2, \quad D_h = \frac{\delta}{2C_h^{\text{win}}}. \end{aligned} \tag{HC3}$$

Here $0 < b < \pi/2$, and $c_h > 0$ is the original proved convolution constant. The source proves $(\omega_{n+r}/\omega_n)^{1/(2r)} \leq C_h^{\text{win}} n$ for $n \geq k \geq 3$, $n/2 \leq r \leq 2n$. Thus it applies to both literal windows, and $0 < D_h < 1/108$.

The exact residual in (HC2) has the following five nonnegative pieces:

$$\begin{aligned} \mathcal{R}_k &= P_- + P_+ + \sum_N w_N \log(1 + \phi_N^2/\mathcal{L}^2) \\ &\quad + \sum_N w_N \log \frac{\varepsilon_N^2 + \phi_N^2}{\mathcal{L}^2 + \phi_N^2} \\ &\quad + 4q \log(C_h^{\text{win}} q) - \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q} + 4q \log \frac{2\mathcal{L}}{\delta k q}. \end{aligned} \tag{HC4}$$

Both sums use exactly $q-1 \leq N \leq 2q-1$. Nonnegativity follows, respectively, from $0 < d_j < 1$, the squared phases, (HT9), the two comparable-window estimates, and $\mathcal{L}/q \geq \delta k/2$. Expanding each logarithm in (HC4), with $D_h = \delta/(2C_h^{\text{win}})$, proves the exact identity

$$\boxed{\mathcal{B}_k = 4q \log(D_h k) + \mathcal{R}_k, \quad \mathcal{R}_k \geq 0.} \tag{HC5}$$

This retains, rather than discards, all the contraction, phase, norm, and trace savings in the original lower bound.

2.2 A proved upper bound for these same four original volumes

The original convolution estimate and Laplace moments are

$$m_{h,k}(u) \geq c_h \vartheta_h^{k-3} e^{-\alpha(|u|+k-3)} (1 + |u| + k - 3)^{-B}, \quad M_h(\pm b) = \int_{\mathbb{R}} e^{\pm bu} w_h(u) du.$$

Write $\mu_h = \int_{\mathbb{R}} w_h(u) du$ for the original source mass. Put

$$\begin{aligned} R_0 &= \sqrt{\delta^2 + \gamma^2}, \quad D = \max(1, 2/b, 2R_0), \quad T = Dq, \quad r = kR_0/T \leq 1/2, \\ a_k(T) &= c_h \vartheta_h^{k-3} e^{-\alpha(T+k-3)} (1 + T + k - 3)^{-B}, \\ M_k &= M_h(b)^k + M_h(-b)^k, \quad U_k(T) = \mu_h^k + \frac{(2q-2)!}{(bT)^{2q-2}} M_k, \\ A_N &= \sqrt{\frac{2N+1}{2}} \frac{4^{N+1} - 1}{3}, \quad B_q(r) = \left(\frac{1+r}{1-r} \right)^q, \quad C_N = \frac{U_k(T) A_N^2 B_q(r)^2}{T a_k(T)}. \end{aligned} \tag{HC6}$$

The capital D in the interval scale is distinct from the inherited lower constant D_h and from the source differential. All masses, coordinates, and root multiplicities remain as in (HT1)–(HT7). One has the actual finite Hermitian comparisons

$$G_{q-1} \preceq C_{2q-1} G_{2q-1}, \quad G_q \preceq G_{q-1} \preceq C_{2q} G_{2q}.$$

Consequently, with the exact ratio

$$\rho_q = \frac{4q+1}{4q-1} \left(\frac{4^{2q+1} - 1}{4^{2q} - 1} \right)^2,$$

$$\boxed{0 \leq \mathcal{B}_k \leq q \log C_{2q-1} + q \log C_{2q} = 2q \log C_{2q-1} + q \log \rho_q.} \tag{HC7}$$

Proof of the degree extension and its maps. Use the exact substitution $f(x) = P(c + iTx)$ with inverse $P(S) = f((S - c)/(iT))$. Its source measure is $Tm_{h,k}(Tx) dx$, with mass μ_h^k , and its monic relation is $\chi_T(x) = (iT)^{-q}\chi(c + iTx)$. The roots $z_1, \dots, z_q$, listed with their repetitions, have modulus at most r . The raw jet of order d acquires exactly $(iT)^d$; the quotient determinant acquires $T^{-q(q-1)}$. This common factor cancels in each displayed quotient-volume ratio.

The Legendre expansion on $[-1, 1]$ gives $\|f\|_{\text{coeff},1} \leq A_N \|f\|_{L^2([-1,1])}$ for $\deg f \leq N$. Indeed Rodrigues' polynomial has coefficient norm at most $\binom{2j}{j} \leq 4^j$ and squared integral $2/(2j+1)$. Pairing the expansion with this polynomial bounds its coefficient by $\sqrt{(2j+1)/2} \|f\|_2$; summing the geometric series gives exactly A_N in (HC6).

Write $\prod_a (1 - z_a t)^{-1} = \sum_{j \geq 0} h_j t^j$. Geometric-series multiplication gives $|h_j| \leq \binom{q+j-1}{j} r^j$ and $\|\chi_T\|_{\text{coeff},1} \leq (1+r)^q$. For every integer $s \geq 0$, monic division of x^{q+s} has quotient $\sum_{j=0}^s h_j x^{s-j}$. The inverse-series identity cancels all other terms of degree at least q . Removing the leading coefficient one therefore gives

$$\|\text{rem}_{\chi_T} x^{q+s}\|_{\text{coeff},1} \leq (1+r)^q \sum_{j=0}^s \binom{q+j-1}{j} r^j - 1 \leq B_q(r).$$

The bound is valid also for $s = q$, the extra degree required here. At degrees below q the remainder is the original monomial, whose coefficient norm is one. Linearity gives the same remainder-map bound for every degree at most $2q$.

For a remainder v of degree below q ,

$$\int |v(u/T)|^2 m_{h,k}(u) du \leq U_k(T) \|v\|_{\text{coeff},1}^2.$$

This follows by bounding each monomial by $\max(1, |u/T|^{q-1})$ and using $\int u^{2q-2} m_{h,k} \leq (2q-2)! b^{-(2q-2)} M_k$. Restriction of the original norm to $[-T, T]$ gives the lower bound $T a_k(T) \|f\|_{L^2([-1,1])}^2$. Combining these three estimates proves

$$\|\text{rem}_\chi P\|_{H_{q-1}}^2 \leq C_N \|P\|_{H_N}^2 \quad (N = 2q-1, 2q).$$

The map is the literal remainder representative of the original jet. Apply it to the canonical least lift of $x \in E$ at degree N . Its norm at degree $q-1$ is at least $x^* G_{q-1} x$; this proves the Hermitian comparisons preceding (HC7). Inclusion of polynomial degrees gives $G_q \preceq G_{q-1}$ and monotonicity of all volumes. Taking determinants proves (HC7). Applying the same remainder inequality to any nonzero polynomial below degree q proves $C_N \geq 1$ without an additional assumption. $\square$

For an entirely scalar upper bound retain the original constants

$$K = \max(1, \vartheta_h^{-1}), \quad C_{\text{env}} = \max(1, 2/(3c_h D)), \quad a = \alpha D + \log 256,$$

and the original expression (AU.31)

$$\mathcal{U}_k = a q^2 + \left(\log(XK) + \alpha + \frac{16R_0}{3D} \right) k q + B q \log(1 + (D+1)q) + q \log C_{\text{env}}.$$

Then the proved four-window upper bound is

$$\boxed{\mathcal{B}_k \leq \widehat{\mathcal{U}}_k := 2\mathcal{U}_k + q \log 20.} \quad (\text{HC8})$$

To verify the constants, $bT \geq 2q$ implies the factorial ratio in (HC6) is at most one; also $\mu_h \leq X$ and $M_k \leq 2X^k$, so $U_k(T) \leq 3X^k$. Directly from (HC6),

$$A_{2q-1}^2 \leq (2q/9)256^q, \quad A_{2q}^2 \leq (40q/9)256^q.$$

The logarithmic derivative $2/(1-r^2) \leq 8/3$ on $[0, 1/2]$ gives $\log B_q(r)^2 \leq 16kR_0/(3D)$. Substitution into C_N yields $q \log C_{2q-1} \leq \mathcal{U}_k$ and $q \log C_{2q} \leq \mathcal{U}_k + q \log 20$. Adding proves (HC8). This argument does not use the false inequality $\rho_2 < 20$; in the actual quartet domain $q \geq 16$ anyway.

2.3 The exact compatible interval for the signed arithmetic correction

Let V_N^Γ denote the inherited Gamma-reference quotient volume for the *same* original polynomial χ , and define

$$T_N = V_N/V_N^\Gamma, \quad \mathcal{B}_k^\Gamma = \log \frac{V_{q-1}^\Gamma V_q^\Gamma}{V_{2q-1}^\Gamma V_{2q}^\Gamma}, \quad \Delta_k^\Gamma = \log \frac{T_{q-1} T_q}{T_{2q-1} T_{2q}}.$$

Cancellation of the four literal volume factors gives $\mathcal{B}_k = \mathcal{B}_k^\Gamma + \Delta_k^\Gamma$. No auxiliary polynomial is introduced and no sign is imposed on Δ_k^Γ . From (HC5) and (HC8) the actual retained residuals must belong to the following explicit intervals:

$$\boxed{\max\{0, -4q \log(D_h k)\} \leq \mathcal{R}_k \leq \widehat{\mathcal{U}}_k - 4q \log(D_h k),} \quad (\text{HC9})$$

$$\boxed{\max\{0, 4q \log(D_h k)\} - \mathcal{B}_k^\Gamma \leq \Delta_k^\Gamma \leq \widehat{\mathcal{U}}_k - \mathcal{B}_k^\Gamma.} \quad (\text{HC10})$$

These intervals have strictly positive width for every $k \geq 3$. Indeed all nonquadratic terms of $\mathcal{U}_k$ are nonnegative, $a > 1$, $q \geq (k+1)^2 > 2k$, and $D_h < 1$. Thus

$$\widehat{\mathcal{U}}_k \geq 2aq^2 > 4qk \geq \max\{0, 4q \log(D_h k)\}.$$

Moreover, keeping the fixed packet constants, each lower-order term divided by q^2 tends to zero, so

$$\lim_{k \rightarrow \infty} \frac{\widehat{\mathcal{U}}_k - \max\{0, 4q \log(D_h k)\}}{q^2} = 2(\alpha D + \log 256) > 0. \quad (\text{HC11})$$

This proves numerical compatibility of the stated actual bound functions. It does not construct a packet realizing an arbitrarily chosen point of either interval, and it does not assert the existence of the hypothetical zero. In particular the interval may lie entirely at negative values of Δ_k^Γ ; determinant-ratio positivity does not imply positivity of its signed four-window logarithm.

2.4 The holonomy energy is tied to the same endpoint contractions

Fix an actual holonomy θ and period L . In this paragraph all metrics, lifts and volumes carry that same phase. Put

$$e_N = \log(V_N/V_{N+1}), \quad \beta_N = \lambda_{\max}(G_{N+1}^{-1/2} G_N G_{N+1}^{-1/2}), \\ J_N^D = \|D_{L,\theta} R_N G_N^{-1/2}\|_{\text{HS}}^2, \quad \mathcal{E}_N = \|\mathcal{B}_N G_N^{-1/2}\|_{\text{HS}}^2.$$

These use original source moments through degree $2N+2$. They obey

$$\boxed{\frac{\mathcal{L}^2}{2} \leq \mathcal{E}_N \leq (1 + \sqrt{\beta_N})^2 J_N^D, \quad e_N \geq 2 \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\}.} \quad (\text{HC12})$$

The denominator is positive: integration by parts on the stated twisted domain gives $\|D_{L,\theta}f\|^2 = c^2\|f\|^2 + \|\partial_r f\|^2$, whence $J_N^D \geq c^2 q > 0$.

To prove the upper inequality, $D_{L,\theta}R_N v$ is an original degree- $(N+1)$ representative of Av . The same quotient minimum therefore gives

$$A^*G_{N+1}A \preceq R_N^*D_{L,\theta}^*D_{L,\theta}R_N.$$

Use $G_N \preceq \beta_N G_{N+1}$, sandwich by $G_N^{-1/2}$, and take traces. This gives $\|R_N A G_N^{-1/2}\|_{\text{HS}}^2 \leq \beta_N J_N^D$. The triangle inequality in $\mathcal{B}_N = D_{L,\theta}R_N - R_N A$ proves the upper inequality in (HC12); (HT14) proves the lower. All eigenvalues in the definition of β_N are at least one by degree monotonicity. In fact the original rank-one kernel update proves the exact identity

$$e_N = \log \beta_N, \quad \beta_N = 1 + \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}}. \quad (\text{HC12a})$$

Here b_{N+1} and ω_{N+1} are the remainder column and monic squared norm for this same original phase measure. Indeed $K_{N+1} = K_N + b_{N+1}b_{N+1}^*/\omega_{N+1}$, so

$$K_N^{-1/2}K_{N+1}K_N^{-1/2} = I + vv^*, \quad v = K_N^{-1/2}b_{N+1}/\sqrt{\omega_{N+1}}.$$

The relative matrix defining β_N is $K_{N+1}^{1/2}K_N^{-1}K_{N+1}^{1/2} = X^*X$ for the invertible $X = K_N^{-1/2}K_{N+1}^{1/2}$. The displayed matrix $I + vv^*$ is XX^* , and $X(X^*X)X^{-1} = XX^*$ proves they have identical eigenvalues. These are 1 with multiplicity at least $q-1$ and $1 + \|v\|^2$. Taking the determinant gives $V_N/V_{N+1} = \det K_{N+1}/\det K_N = 1 + \|v\|^2 = \beta_N$, which proves (HC12a), including a zero update if one occurs. Rearranging the two energy bounds now proves the second inequality of (HC12) with this exact endpoint identity.

The same weights as in (HC2) reconstruct the actual phase volume:

$$\mathcal{B}_{k,\theta} = \sum_{N=q-1}^{2q-1} w_N e_N \geq 2 \sum_{N=q-1}^{2q-1} w_N \log \max \left\{ 1, \frac{\mathcal{L}}{\sqrt{2J_N^D}} - 1 \right\}. \quad (\text{HC13})$$

Thus a small original derivative-source energy at a degree has the explicit endpoint contraction cost (HC12); that cost enters the same four-window budget, with its actual multiplicity.

Finally the supplied holonomy construction allows any fixed $0 < \eta < 1$ by choosing the original period from its computed weighted source comparison at degree $2q+2$. It proves simultaneously $(1-\eta)G_N \preceq G_{N,\theta} \preceq (1+\eta)G_N$ at all these degrees. Taking determinants at the four original endpoints gives

$$|\mathcal{B}_{k,\theta} - \mathcal{B}_k| \leq 2q \log \frac{1+\eta}{1-\eta}. \quad (\text{HC14})$$

Consequently (HC13)'s right side is at most $\widehat{\mathcal{U}}_k + 2q \log((1+\eta)/(1-\eta))$. For the averaged original phase metrics the already proved stronger comparison $(1-\eta^2)G_N \preceq \overline{G}_N \preceq G_N$ gives instead $|\overline{\mathcal{B}}_k - \mathcal{B}_k| \leq 2q \log(1/(1-\eta^2))$. These changes are of order q for a fixed η , whereas the compatible width (HC11) has positive quadratic order. The exact Laplacian residual (HT17) remains present at every phase. No displayed upper estimate for its norm, or for the original derivative moments in (HC12), contradicts these lower bounds.

2.5 The sharper finite comparison retains its exact error

The original lower-envelope measure is

$$d\nu_k(u) = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} e^{-\alpha|u|} (k-2+|u|)^{-B} du.$$

Let V_N^ν be its quotient volume for the identical full remainder map. The actual finite upper expression for the four-window volume is

$$\mathcal{U}_{\nu,k} = \log(V_{q-1} V_q) - \log(V_{2q-1}^\nu V_{2q}^\nu),$$

with the exact nonnegative error

$$\boxed{\mathcal{U}_{\nu,k} - \mathcal{B}_k = \log \frac{V_{2q-1}}{V_{2q-1}^\nu} + \log \frac{V_{2q}}{V_{2q}^\nu} \geq 0.} \tag{HC15}$$

Indeed $m_{h,k}(u) du \geq d\nu_k$ bounds every source polynomial quadratic form. Minimizing on each identical affine remainder fibre gives $G_N \succeq G_N^\nu \succ 0$. Conjugation by $(G_N^\nu)^{-1/2}$ shows that each volume ratio in (HC15) is at least one. Cancellation proves the equality. The low-degree factors require only original moments through $2q$, and the comparison moments in the last factors go through $4q$. These are well-defined finite expressions, with the actual full jet map and source mass. The supplied proof records do not evaluate their errors at the hypothetical arithmetic quartet or bound them at a rate opposing (HC5). Retaining this sharper finite expression therefore supplies no extra contradictory inequality.